

1. Consider the Robertson-Walker spacetime described by the metric

$$ds^2 = dt^2 - e^{2t/t_H} [dr^2 + r^2(d\theta^2 + \sin^2 \theta d\phi^2)] , \quad (1)$$

with the coordinates $x^\mu = (t, r, \theta, \phi)$, where t is the universal time and $t_H \approx 14$ Gyr is the Hubble time.

- (a) (5 points) Compute the path $r(t)$ of a light pulse emitted at time $t = t_0$ at the origin $r = 0$. You may assume that the light ray moves along the line $\theta = \pi/2$ and $\phi = 0$ so that $dr/dt > 0$.
- (b) (5 points) Compute the proper distance between the origin $r = 0$ and a point $r > 0$ on the line $\theta = \pi/2$ and $\phi = 0$ at a fixed universal time t .
- (c) (5 points) Compute the spectral shift of the light pulse in a) during the time between emission at the origin and arrival at a point $r > 0$.

Hint: the spectral shift is defined as $z = \lambda_{rec}/\lambda_{em} - 1$, where λ_{em} is the wavelength of the light at emission and λ_{rec} its wavelength when it arrives.

2. A scalar field ϕ with potential energy density $V(\phi)$ has a Lagrangian density given by

$$\mathcal{L} = \frac{1}{2} g^{\mu\nu} (\partial_\mu \phi) (\partial_\nu \phi) - V(\phi) . \quad (2)$$

- (a) (5 points) Derive the equation of motion for the scalar field ϕ .
- (b) (10 points) Assuming that the $N + 1$ -dimensional spacetime has a metric given by

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu = dt^2 - a(t)^2 G_{ij} dx^i dx^j , \quad (3)$$

where G_{ij} are the metric components on an N -dimensional Riemannian manifold, and that the scalar field ϕ only depends on the time coordinate t , show that the scalar field is an ideal fluid and find the (time-independent) equation-of-state parameter $w = p/\rho_0$.